

CASE STUDY

Learning Game Development

Designing a game-based learning experience with custom Storyline development

ROLE Learning Technologist / Learning Experience Developer / eLearning Developer / Web Designer	PRIMARY TOOLS Storyline • JavaScript • PHP/MySQL	FOCUS Gamification • Game Mechanics • Leaderboard
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The screenshot shows a web browser window with the URL `C:/Users/KHaas/OneDrive%20-%20Sodexo/Desktop/Sodexo/Projects/Archive/GLOBAL-1/Game/STORYL-1/FMOPPO-1/FMOPPO-1/STORY-1.HTM`. The game interface is titled "Round 1 of 4: Grounds Maintenance". A central text box instructs the player: "For this round, only click on indicators that suggest a potential Grounds Maintenance FM opportunity." The game environment is a 3D-rendered courtyard with a building in the background, trees, and a path. A "You are here" map is located in the bottom left corner. The bottom of the screen features a control panel with a progress bar for "Opportunity 1 of 4" and "Indicators 0 of 8", a "Your Score" display showing 0, a "Time Remaining" display showing 3:59, and buttons for "Help", "I Give Up!", and "Save Game".

[Click here to view the full game](#)

The Challenge

The facility management training program needed a learning experience that went beyond a traditional linear, click-and-read course. The goal was to create a performance-based game that allowed learners to explore a realistic environment, make decisions, receive feedback, and apply their knowledge while also introducing friendly competition.

Our Approach

I co-developed the game with a colleague. We worked together on the overall experience design and game mechanics. My colleague focused primarily on the graphic design and user interface, while I focused on development in Articulate Storyline and building the companion leaderboard. There was a lot of collaboration between the two of us throughout the entire process, including bouncing ideas off each other and fine-tuning and iterating to improve the game experience.

The game was part of a blended learning program that combined the digital experience with live facilitated training. The design emphasized exploration, contextual decision-making, scoring, time pressure, progress indicators, and feedback. These elements were intended to make the learning experience feel more like a real task than a conventional course.

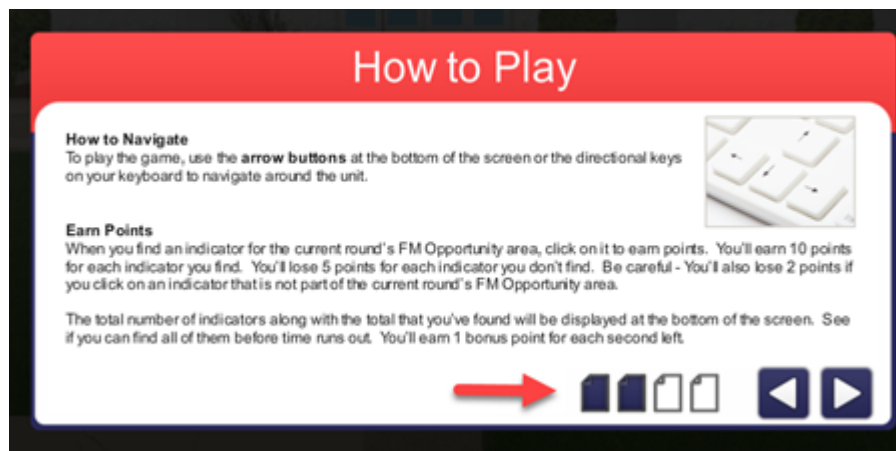
Game Design & Development

ARTICULATE STORYLINE Built the core game experience, including custom navigation, movement, scoring, timing, progress indicators, feedback, and game logic.	GAME MECHANICS Co-developed exploration, distractors, point scoring, time pressure, contextual navigation, feedback, and the “I give up” mechanic.
GRAPHIC DESIGN / UI Planned and brainstormed the overall visual direction with my colleague, who led the graphic design and user-interface work.	LEADERBOARD DEVELOPMENT Built the companion leaderboard using HTML5, CSS, JavaScript, jQuery, PHP, and MySQL to store and display player scores.

Interactive Game Mechanics

Several game mechanics were used to create the game experience:

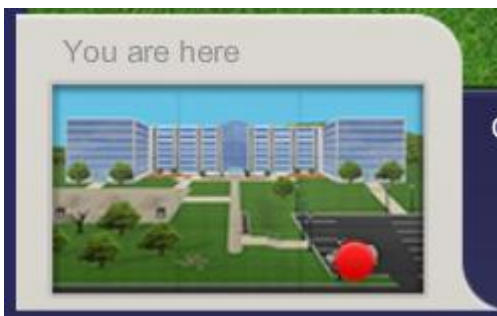
A gate screen provided starting instructions:



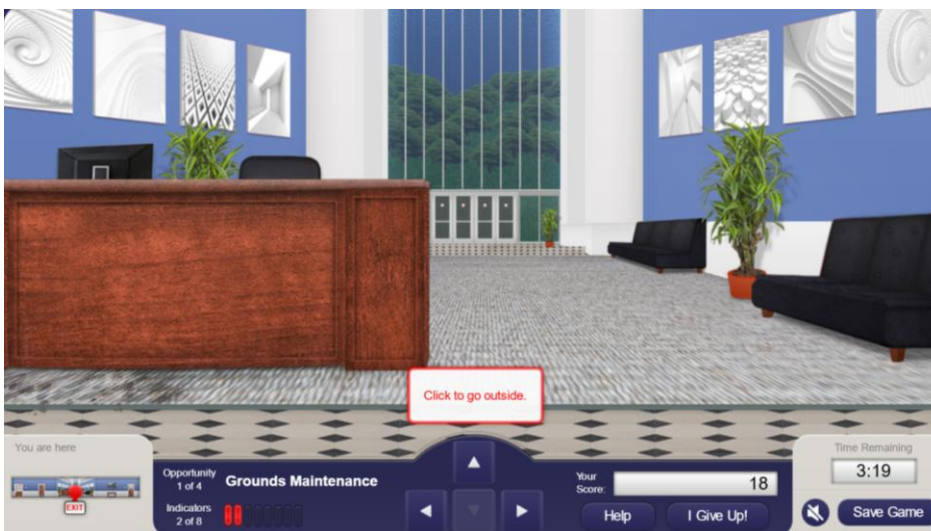
Custom navigation allowed learners to navigate the game environment using the up/down/left/right buttons or the arrow keys on their keyboard. Learners explored the environment and made educated guesses, with specific feedback. Distractors and incorrect choices were intentionally included so learners had to focus on the task and context. A timer was used to create a sense of urgency. Progress indicators portrayed game progress. A point system was used to provide friendly competition.



A contextual map showed where you are within the game environment:



There were multiple areas/rooms within the game environment to explore:



A summary and score breakdown was provided after each of the four rounds of the game:

FM

FM Opportunities Challenge: Virtual Search Game

Results

Round 1: Grounds Maintenance

Sorry, you ran out of time!

	round points ▼
Indicators Found (+10 each):	2 of 8 = 20
Indicators Not Found (-5 each):	6 of 8 = -30
Indicators Found Wrong Round (-2 each):	1 = -2
Time Bonus Points (+1 each second):	0:00 left = 0
Grounds Maintenance Points:	-12

Total Points:

1

x

-12

The indicators you found suggest a potential opportunity to offer Grounds Maintenance services at this unit. However, there are many factors to consider when evaluating whether this opportunity is actually worth pursuing. In this game you're only focusing on identifying potential opportunities, but later on in the Global FM Academy training you'll learn how to properly assess whether an opportunity like this is worth acting on.

Note: At the end of the game, you'll be given the opportunity to review each indicator you found or did not find.

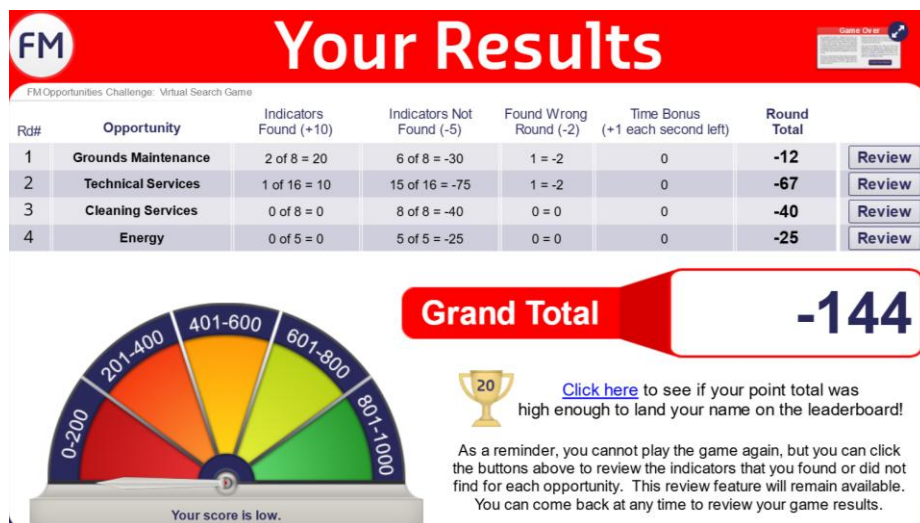
Next Up: Round 2 Technical Services

Next, you'll work on finding indicators for a potential Technical Services FM opportunity at this unit. You'll begin at the same location within the unit as where you left off in the prior opportunity. Let's see how many of the 16 indicators you can find. You have 8 minutes.

Click **Continue** when you're ready!

Continue

At the end of the game a breakdown of your final score was provided:



Leaderboard & Technical Implementation

The companion leaderboard extended the learning experience beyond the Storyline file. I built the leaderboard website with HTML5, CSS, JavaScript, jQuery, PHP, and MySQL. A JavaScript trigger sent player data to the MySQL database so scores could be stored and retrieved for friendly competition across the organization. This gave the game a persistent competitive element outside the course itself.

STORYLINE	JAVASCRIPT	PHP	MYSQL	LEADERBOARD
Game & scoring	Player data	Server logic	Score storage	Competition

FM Opportunities Challenge Virtual Search Game English

Total # of Players: 1,458

Top 20: All Time Highest Scores

#	User	Date	Score
1	User 1	9/22/2015	2,956
2	User 2	9/22/2015	1,936
3	User 3	9/22/2015	1,856
4	User 4	9/22/2015	1,736

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Recognition & Impact

We showcased the game at the Learning Solutions 2016 Conference DemoFest and presented about it in a hands-on session at DevLearn 2016. Articulate later [featured the game](#) in their Rapid E-Learning Blog.

DEMOFEST Learning Solutions 2016	DEVLEARN 2016 presentation (hands-on session)	ARTICULATE Featured in Rapid E-Learning Blog
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What I Learned

This project reinforced the value of combining learning design, game mechanics, visual design, and custom development. It demonstrated how Storyline could support a highly interactive, game-based learning experience when paired with variables, triggers, motion paths, JavaScript, and a separate web-based leaderboard.

[Click here to view the full game](#)